In [ ]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_sc
import shap


df = pd.read_csv('/content/heart_attack_prediction_dataset.csv')
```

In [ ]:
```
df
```

Out[ ]:

	Patient ID	Age	Sex	Cholesterol	Blood Pressure	Heart Rate	Diabetes	Family History	Smokin
0	BMW7812	67	Male	208	158/88	72	0	0	
1	CZE1114	21	Male	389	165/93	98	1	1	
2	BNI9906	21	Female	324	174/99	72	1	0	
3	JLN3497	84	Male	383	163/100	73	1	1	
4	GFO8847	66	Male	318	91/88	93	1	1	
...	...	...	...	...	...	...	...	...	
8758	MSV9918	60	Male	121	94/76	61	1	1	
8759	QSV6764	28	Female	120	157/102	73	1	0	
8760	XKA5925	47	Male	250	161/75	105	0	1	
8761	EPE6801	36	Male	178	119/67	60	1	0	
8762	ZWN9666	25	Female	356	138/67	75	1	1	

8763 rows × 26 columns

In [ ]:
```
df['Sex'] = df['Sex'].map({'Male': 1, 'Female': 0})
df['Diet'] = df['Diet'].map({'Healthy': 2, 'Average': 1, 'Unhealthy': 0})

bp_split = df['Blood Pressure'].str.split('/', expand=True)
df['BP_Systolic'] = bp_split[0].astype(float)
df['BP_Diastolic'] = bp_split[1].astype(float)
```

```
df.drop(['Blood Pressure'], axis=1, inplace=True)

df.drop(columns=['Patient ID', 'Country', 'Continent', 'Hemisphere'], inplace=Tr

X = df.drop('Heart Attack Risk', axis=1)
y = df['Heart Attack Risk']
```

In [ ]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3,
```

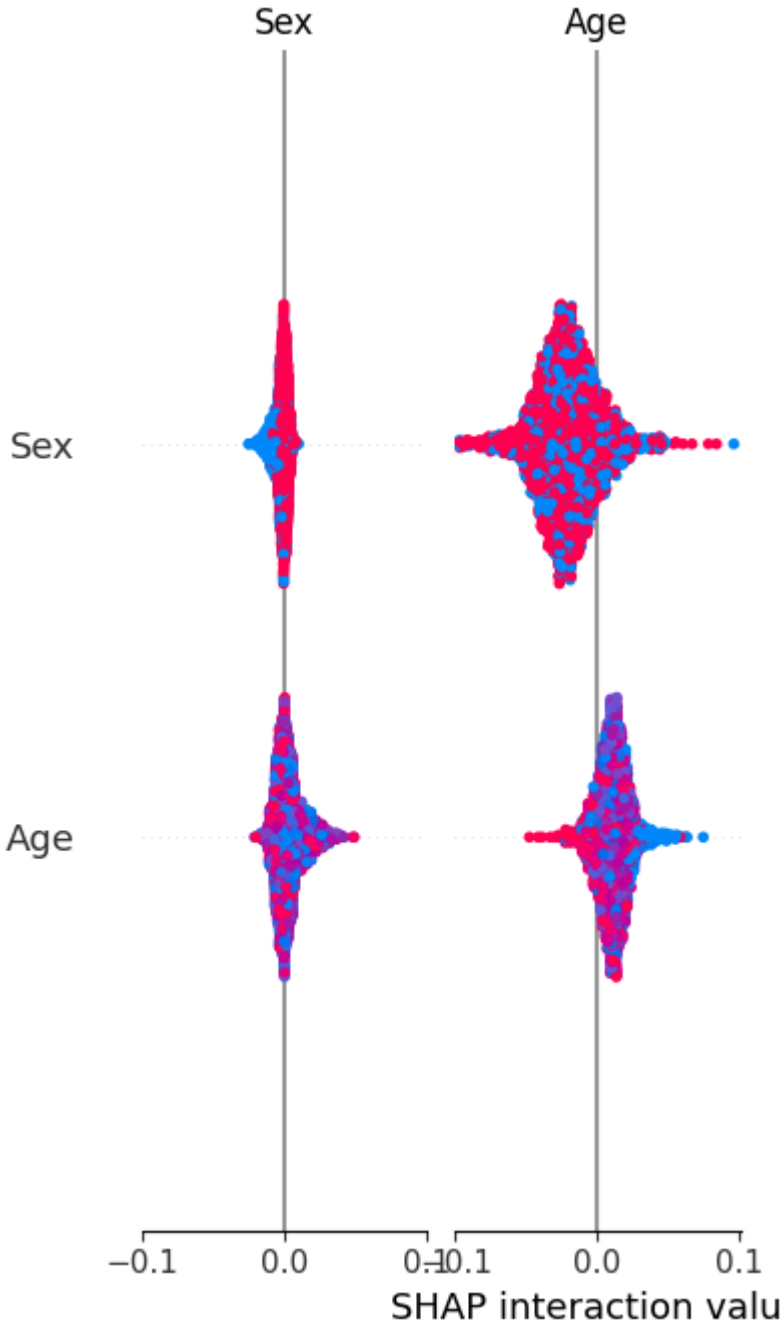
In [ ]:
```
clf = RandomForestClassifier(n_estimators=100, random_state=42, class_weight='ba
clf.fit(X_train, y_train)
```

Out[ ]:
```
RandomForestClassifier(class_weight='balanced', random_state=42)
```

In [ ]:
```
y_pred = clf.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.6394066184861164
Confusion Matrix:
 [[1667   24]
 [ 924   14]]
Classification Report:
              precision    recall  f1-score   support

           0       0.64      0.99      0.78      1691
           1       0.37      0.01      0.03       938

    accuracy                           0.64      2629
   macro avg       0.51      0.50      0.40      2629
weighted avg       0.55      0.64      0.51      2629
```

In [ ]:
```
# SHAP explanation
explainer = shap.TreeExplainer(clf)
shap_values = explainer.shap_values(X_test)
shap.summary_plot(shap_values, X_test, feature_names=X.columns)
```


Sex
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SHAP interaction valu